

Affordance Maintenance-based 3D Scene Synthesis for Immersive Mixed Reality

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ABSTRACT

With the explosion of head-mounted displays into the commercial market, immersive scenes are being used in many applications, *e.g.*, training and entertainment. As these applications are usually used indoors, the user would suffer from safety and immersion issues when immersed in a virtual scene. Therefore, we propose a 3D synthesis method that synthesizes the 3D scene according to the affordance of physical objects in the situated physical environment and the intention of the user. The results show that virtual objects in synthetic scenes have the same affordances as physical objects in the real world and can provide interaction medium and feedback to the situated user and avoid collisions.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality; Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality

1 INTRODUCTION

Virtual scenes can provide a highly interactive and immersive environment for the user in many applications (*e.g.*, games, training, and education). Most VR applications are used in indoor environments. Physical objects could provide their function and passive feedback for immersive users, but there is a risk that the user in VR will clash with the physical objects [5]. Previous work has shown that taking advantage of the affordance provided by the physical objects for situated VR experience could improve user experience and task performance [1] and synthesize virtual scenes based on physical environments [6]. However, these works only focused on specific scene synthesis using similar virtual objects as physical objects or only considered the walk area [6] without or with one or two interactive physical objects.

Therefore, one of our goals is to synthesize virtual scenes for users based on their situated physical environments, considering the affordance of the physical objects. In addition to physical information, we consider the user's intention which is expressed by the text or speech which can be easily converted into text through the speech-to-text technique. In particular, the texts from users are unstructured. Large language models (LLMs) can effectively extract information from unstructured text (*e.g.*, entities, relationships, and events). Based on an LLM model, We propose an agent, an inference model, that can observe the user's intentions and the user's situated physical environments and then synthesize virtual scenes with interactivity as shown in figure 1. For each physical object, the proposed agent predicts the affordance similarity probability and size similarity probability between the virtual object and the physical object and predicts the place, season, and object similarity probability between the virtual object and the input text from the user, which are used to infer the virtual object corresponding the physical object. Finally, according to the physical information, the world agent synthesizes scenes by translating, rotating, and scaling the virtual object. The synthesized results show that the synthesized scene can not only maintain the same affordances as physical objects but also maintain the style of the virtual environment that the user wants.

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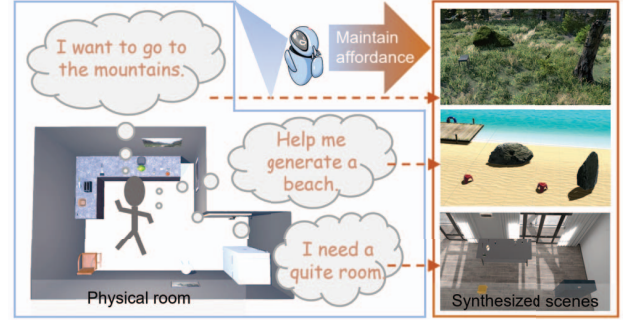


Figure 1: The scenes synthesized by the proposed agent maintain the affordance of the physical environment and the style mentioned by users.

2 SYNTHESIZED METHOD

Based on the physical information and the user input, the agent predicts the virtual object for each physical object. Figure 2 shows the method - the selection pipeline of each virtual object for the corresponding physical objects. Finally, all selected virtual objects are scaled according to the size of the corresponding physical objects, and the virtual objects are placed according to the position and rotation of the real objects to synthesize the virtual scenes.

Preliminary. Based on the observation $C = \{\mathcal{T}, \mathcal{O}^r\}$, the agent synthesizes virtual scene $\mathcal{S}$. $\mathcal{T}$ is the input text prompt from users to describe the scene they want to. $\mathcal{O}^r$ is the information of the real world users are situated in, where $\mathcal{O}^r = \{o_i^r\}_{i=1}^N$ and each tuple o_i^r denotes the information of a physical object. $\mathcal{S}$ is the virtual scene users want to synthesize, where $\mathcal{S} = \mathcal{O}^v = \{o_j^v\}_{j=1}^M$, where $\mathcal{O}^v$ is the set of all virtual objects and each tuple o_j^v denotes the information of a virtual object. Object information includes a category, description of the object (can be empty), bounding box location, bounding box rotation, and bounding box size. $\mathcal{A} = \{a_k\}_{k=1}^K$ is a tuple of K kind of affordances of objects, where each tuple a_k denotes one kind of affordance. We consider ten different affordances for the objects including *walkable*, *walkable*, *supportable*, *sitable*, *drinkable*, *eatable*, *graspable*, *breakable*, *dangerous*, *moveable*, *obstruct*. $\mathcal{P} = \{place_k\}_{k=1}^K$ is a tuple of K kind of virtual places users usually want to go, where each tuple $place_k$ denotes the label of a virtual place. we consider the probability that each object could appear in different places including *Library*, *Conservatory*, *Spa*, *Lounge*, *Observatory*, *Suite*, *Monastery*, *Studio*, *Bookstore*, *Aquarium*, *Beach*, *Forest*, *Garden*, *Vineyard*, *Yacht*, *Rooftop*, *Treehouse*, *Reef*, *Peak*, *Rainforest*. $\mathcal{E} = \{e_k\}_{k=1}^K$ is a tuple of K kind of seasons, where each tuple e_k denotes one kind of season. We consider the probability that each object could appear in each reason including *spring*, *summer*, *autumn*, *winter*.

Virtual object-based inference. For each virtual object, the LLM model predicts the likelihood for each affordance, place, and season in the list which are represented by one-dimensional vectors respectively as:

$$V(\mathcal{A}|o_j^v), V(\mathcal{P}|o_j^v), V(\mathcal{E}|o_j^v) = f_{LLM}(o_j^v) \quad (1)$$

User intention inference. Given the text prompt $\mathcal{T}$, the virtual place and season the user wants to go and the virtual objects the user wants are extracted by the LLM model:

$$intent_place, intent_season, intent_objects = f_{LLM}(\mathcal{T}) \quad (2)$$

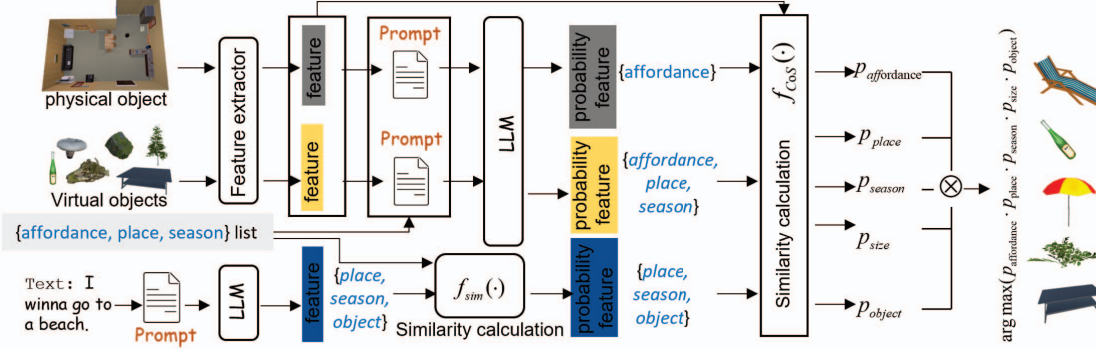


Figure 2: The synthesis algorithm. The affordance attribute probabilities of each physical object are predicted by an LLM model; the attribute probabilities of virtual objects are predicted by the LLM model; the text features are extracted by the LLM model, which is used to predict the similarity with all places, season and virtual object attribute probabilities via a similarity calculation module. Then, another similarity calculation module is used to predict the five similar probabilities: the affordance similarity probabilities between the physical object and all virtual objects; the place, season, and object similarity probabilities between the virtual objects and the user input. Finally, the corresponding virtual object consistent with the physical object is predicted.



Figure 3: The synthesized scenes.

A similarity prediction module Sim [3] predicts the the similarity between the user-mentioned $intent_place$, $intent_season$, and $intent_objects$ and defined lists of places $\mathcal{P}$, $\mathcal{E}$ and O^v as one-dimensional vectors respectively as:

$$\begin{aligned} V(\mathcal{P}|intent_place) &= f_{Sim}(\mathcal{P}, intent_place) \\ V(\mathcal{E}|intent_season) &= f_{Sim}(\mathcal{E}, intent_season) \\ V(O^v|intent_objects) &= f_{Sim}(O^v, intent_objects) \end{aligned} \quad (3)$$

Physical object-based inference. For synthesizing scenes where users could interact by taking advantage of passive feedback from physical objects, the affordance of the virtual object should be aligned with the corresponding physical object. Given a physical object o_i^r , the likelihood of each affordance for it is predicted by LLM model which is represented by one-dimensional vectors respectively as:

$$V(\mathcal{A}|o_i^r) = f_{LLM}(o_i^r) \quad (4)$$

Joint inference. The affordance similarity between a virtual object o_j^v and a physical object o_i^r is measured by distances with cosine similarity CoS as:

$$p_{aff} = f_{Cos}(V(\mathcal{A}|o_i^r), V(\mathcal{A}|o_j^v)) \quad (5)$$

The place, season and object similarities between a virtual object o_j^v and the user intention is defined as:

$$p_{place} = f_{Cos}(V(\mathcal{P}|o_j^v), V(\mathcal{P}|intent_place)) \quad (6)$$

$$p_{season} = f_{Cos}(V(\mathcal{E}|o_j^v), V(\mathcal{E}|intent_season)) \quad (7)$$

$$p_{object} = f_{Cos}(V(\mathcal{O}|o_j^v), V(O^v|intent_objects)) \quad (8)$$

Finally, the virtual object should have a similar size to the corresponding physical object. The size similarity between a virtual object o_j^v and a physical object o_i^r is represented as p_{size} . If $0.5 * size_{o_i^r} \leq size_{o_j^v} \leq 2.0 * size_{o_i^r}$, $p_{size} = 1.0$, else $p_{size} = 0.1$.

For each physical object o_i^r , the world agent predicts the probability that it maps to each virtual object o_j^v as:

$$p(o_j^v|\mathcal{T}, o_i^r) = p_{aff} * p_{place} * p_{season} * p_{object} * p_{size} \quad (9)$$

Given an observation C , synthesizing the virtual scene S is to find the optimal corresponding virtual object o_j^v from O^v for each

physical object o_i^r . The likelihood of synthesizing a virtual scene S can be written as:

$$p(S|C) = p(S|\mathcal{T}, \mathcal{R}) = \prod_{i=1}^N \argmax_{o_j^v \in O^v} p(o_j^v|\mathcal{T}, o_i^r) \quad (10)$$

3 RESULTS AND CONCLUSION

We synthesized virtual scenes based on the physical rooms from the ProcTHOR [2] and ScenenNN [4] datasets. Figure 3 shows four synthesized virtual scenes, where when the physical rooms are different, the synthesized virtual scenes under the same user intention are different, and for the same physical environment, the synthesized virtual scenes under different user intentions are different. The synthesized results show that the synthesized scenes maintain affordances as physical objects and the style that the user wants. In the future, we will synthesize virtual scenes based on real physical rooms, compare our methods with other methods, and conduct user studies to test the user's sense of security, immersion, and interactivity of objects to evaluate the proposed method.

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